

Person 2 – Buffer Analysis and Exclusion Areas

Methods for the Analysis and Modelling of Spatial Environmental Data — Assignment 4, Summer Semester 2026

Study area: ~182 km² west of Hanover | CRS: EPSG:31467 (DHDN / Gauss-Krüger zone 3)

1. What this stage of the project was about

Person 1 handed over the QGIS project with the seven original shapefiles, the georeferenced aerial photo, and the digitized oxbow arms of the Leine river. Our job was to turn all of that into one thing: a single layer showing every area where a wind turbine legally cannot be built, based on the minimum distances given in the assignment. This means building eleven separate safety buffers (roads, railway, power lines, water, forest, oxbow arms, and settlements) and then merging them into one clean exclusion area that Person 3 can subtract from the usable farmland.

2. The eleven buffers

Each buffer distance and the exact selection used is listed below. Straßennetz.shp holds several road categories in one field (typ); we only buffered the three categories the assignment actually asks for and left the rest (Gemeindestraße, Weg, Straße sonstige) untouched, since applying a distance to them isn't part of the criteria.

Feature	Buffer	Selection used
District roads (Kreisstraße)	15 m	typ = 'Kreisstraße'
Federal / state roads	20 m	typ IN ('Bundesstraße','Landesstraße, Staatsstraße')
Motorways (Bundesautobahn)	40 m	typ = 'Bundesautobahn'
Railway lines	40 m	all features
Overhead power lines	195 m	all features (3 × 65 m rotor diameter)
Water bodies	25 m	all features
Oxbow arms (Altwasser)	200 m	all features (protected biotope)
Forest	200 m	typ = 'Wald; Forst'
Settlement / industrial / commercial / special-function areas	650 m	Siedlungsfläche intersecting Ortslagen, or typ IN ('Indus.-Gew.-Fl.','Fl. fkt. Praeg.')
Individual houses / settlement splinters	450 m	Siedlungsfläche NOT intersecting Ortslagen

All buffers were run with the same settings for consistency: 16 segments per quarter-circle, round caps and joins, and 'dissolve result' switched on so overlapping parts of the same class merge automatically.

3. The settlement problem, and how we solved it

This was the one part that wasn't straightforward. The assignment separates 'settlement and residential areas' (650 m) from 'individual houses and settlement splinters' (450 m), and defines the splinters as settlement areas 'not assigned to a local location'. The trouble is that Landnutzung.shp has no field that marks this difference — every settlement patch just carries the same value, Siedlungsfläche.

We solved it with a spatial comparison instead of an attribute filter: any Siedlungsflaeche polygon that touches an Ortslagen polygon (the layer of named villages) counts as a settlement assigned to a local location, and gets the 650 m buffer along with industrial, commercial and special-function areas. Anything that doesn't touch Ortslagen — isolated farms, hamlets, scattered houses — gets the 450 m buffer instead. Out of 1,771 Siedlungsflaeche polygons, 1,640 (about 3,231 ha) are attached to a named village, and 131 (about 118 ha) sit on their own. This matches the assignment's own wording almost exactly, so we're confident it's the right interpretation, not just a convenient one.

4. Putting the eleven buffers together

Once all eleven buffers existed, we combined them into one exclusion layer in four steps:

- Merge – all 11 buffer layers combined into a single layer.
- Fix geometries – cleaned up before dissolving, not after. Merging that many overlapping polygons can leave small invalid slivers, and dissolving on top of those can create gaps or broken multipart features that a later 'fix' won't undo. Repairing first keeps the topology clean going into the dissolve.
- Dissolve – no grouping field, so every buffer that touches another one melts into a single shape with no leftover internal lines.
- Clip – cut to the Frame boundary, so buffers that stuck out past the edge of the study area don't count.

The result, Total_Exclusion_Area, is exactly one polygon feature.

5. How much area is actually excluded

This is where it's worth being precise rather than using the assignment's rounded '~200 km²'. The Frame boundary measures 181.78 km² (18,177.8 ha) — that's the real, measured figure, not an estimate. The exclusion area comes out to 17,519.48 ha.

That means about 96.4% of the study area is off-limits once every buffer is applied, leaving roughly 658 ha (6.6 km²) as scattered, unconstrained patches. That's a small remainder, but it's not a mistake — the 650 m settlement buffers are large and this is a fairly densely settled part of Lower Saxony, so they overlap heavily and swallow most of the map. You can see this visually too: the exclusion layer looks almost solid, with only small isolated white 'holes' left over, and those holes are exactly the candidate zones Person 3 needs to filter further.

6. A few things that went wrong along the way (and how we caught them)

Worth mentioning briefly, since it explains some of the choices above:

- Early attempts at extracting Wald;Forst using IN ('Wald','Forst') matched nothing, because the field actually stores the combined string 'Wald; Forst' — the semicolon is part of the value, not a separator.
- We tested an extra safety filter ($\$area < 10,000,000$) to screen out potential oversized outlier polygons before buffering forest and settlement classes. It turned out to do nothing: the largest forest polygon is 261.8 ha and the largest Siedlungsflaeche polygon is 22.0 ha, both far below that threshold. We removed it from the final expressions rather than keep a condition that has no real effect.

- A couple of intermediate results were briefly left as temporary, in-memory layers before being properly saved to file — fixed by exporting each one explicitly before moving to the next step.

7. What we're handing to Person 3

- Total_Exclusion_Area.gpkg — the final exclusion layer (main deliverable).
- Buffer.gpkg — all eleven individual buffers plus the merged and dissolved intermediate layers, kept for traceability.
- Altwasser.gpkg and Luftbild_georef.tif — carried over from Person 1.

Next step for Person 3: take the land-use classes Ackerland and Grünland, subtract Total_Exclusion_Area using a Difference operation, convert the result from multipart to singlepart, calculate area in hectares, and keep only the patches of at least 18 ha — enough room for three wind turbines.